

ICSA 2026 SHORT COURSE

Lecture 4: Posterior drift and biased regularization

Statistical Foundations of Transfer Learning

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Overview

§4.1: Posterior drift

§4.2: Biased regularization

§4.2.3: Adaptive ℓ_2 -penalty (Duan et al.)

Numerical illustration: biased ridge

§4.3: High-dimensional extensions

§4.3.3: Block penalty with multi-task learning

§4.3.4: Shared low-dimensional representations

§4.4: Unsupervised federated transfer

§4.5: Federated transfer + differential privacy

Lecture summary

References

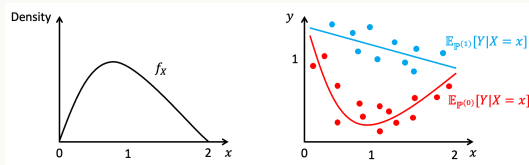
§4.1: Posterior drift

Posterior drift, formally

L3 corrected a moving $\mathbb{P}_X^{()}$ by reweighting; now $\mathbb{P}_X^{()}$ is shared but the labels drift, so reweighting cannot help — we shrink toward the source instead.

- Source $(X, Y) \sim \mathbb{P}^{(1)}$, target $(X, Y) \sim \mathbb{P}^{(0)}$.
- **Posterior drift:** $\mathbb{P}_X^{(0)} = \mathbb{P}_X^{(1)}$, $\mathbb{P}_{Y|X}^{(0)} \neq \mathbb{P}_{Y|X}^{(1)}$.
- Same features, different conditional model: the regression / decision function moves while X -marginal is fixed.

Running example: same scanner ($\mathbb{P}_X^{()}$ shared), but two hospitals label borderline cases differently — posterior drift in our L1 MRI story.



Posterior drift vs. concept drift

	$\mathbb{P}_X^{()}$ shift?	$\mathbb{P}_{Y X}^{()}$ shift?
Covariate shift (L3)	Yes	No
Posterior drift (L4)	No	Yes
Concept drift	Yes	Yes

Why this matters: under posterior drift, the source's X -distribution is uninformative about *which* target points are hard. All the structure to transfer lives in the conditional law itself.

Setup for the rest of the lecture

Linear / GLM regression, source coefficient $\beta^{(1)*}$, target coefficient $\beta^{(0)*}$.

- Source: $y_i^{(1)} = \mathbf{x}_i^{(1)\top} \beta^{(1)*} + \varepsilon_i$, $i = 1, \dots, n_1$ (large).
- Target: $y_i^{(0)} = \mathbf{x}_i^{(0)\top} \beta^{(0)*} + \varepsilon_i$, $i = 1, \dots, n_0$ (small).
- Drift parameter $\delta := \beta^{(0)*} - \beta^{(1)*}$, hopefully small in some norm.

Today's question: *when does using the source help estimate $\beta^{(0)*}$?*

Three things we'd like

Analogy: a new doctor using a related hospital's evidence — learn from it when patients are similar (1), ignore it when populations differ (2), and not need a consultant to know which regime you're in (3).

1. **Improvement when source is close.** If $\|\delta\|$ is small, transfer should beat target-only.
2. **Safety when source is far.** If $\|\delta\|$ is large, we want to *avoid negative transfer* (*negative transfer*: using the source makes the target estimator *worse* than ignoring it): the transfer estimator shouldn't be much worse than target-only.
3. **Adaptivity.** Without knowing $\|\delta\|$ in advance, the procedure should pick its own degree of borrowing.

These three are the through-line of the lecture: *biased regularization* delivers (1)+(2); *adaptive penalty selection* delivers (3).

Check-in: setup

*Under pure posterior drift, which of the following is **true**?*

- (a) Importance-weighted ERM (Lecture 3) is the right tool.
- (b) Source data and target data are exchangeable, so just pool them.
- (c) Source-only ERM is permanently biased on target, but *any* target data plus source bias regularization can help.
- (d) No method using source data can help; transfer is impossible.

Answer

(c). P_X identical $\Rightarrow$ reweighting (a) cannot help; pooling (b) is biased since conditionals differ; (d) is wrong — small $\|\delta\|$ makes source valuable. Only (c) names the right tool.

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§4.2: Biased regularization

Why biased regularization?

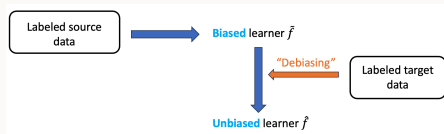
Standard ridge: $\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda \|\beta\|^2$ shrinks toward $\mathbf{0}$.

Biased ridge: shrink toward the *source* estimate $\hat{\beta}^{(1)}$:

$$\hat{\beta}_{\text{bias}}^{(0)} = \arg \min_{\beta} \|\mathbf{y}_T - \mathbf{X}_T \beta\|^2 + \lambda \|\beta - \hat{\beta}^{(1)}\|^2.$$

*Think of it as a **leash anchored to the source**: λ sets leash length, and the optimal length shrinks as the source gets farther.*

We previewed this in L2 (the weighted-pooling bound); here we develop it via biased regularization, where L2's per-domain weights become the penalty λ .



Application examples

- **Fine-tuning** a pretrained model: weight-decay toward pretrained weights $\hat{\beta}_{\text{pre}}$ is biased ridge with $\hat{\beta}^{(1)} = \hat{\beta}_{\text{pre}}$.
- **Multi-site clinical trials**: each site shrinks toward an aggregate prior, recovering the same ℓ_2 -penalty story.
- **Personalized models**: each user's model shrinks toward a population-level coefficient.
- **Bayesian view**: $\lambda \|\beta - \hat{\beta}^{(1)}\|^2$ is the negative log-prior $\beta \sim \mathcal{N}(\hat{\beta}^{(1)}, \frac{\sigma^2}{\lambda} I)$.

Biased ridge: exact bias-variance decomposition

Take $\mathbf{X}_T^\top \mathbf{X}_T = n_0 \cdot \mathbf{I}_p$ (orthonormal design) and σ^2 -noise. Then with anchor $\hat{\boldsymbol{\beta}}^{(1)}$, (*subscript T denotes the target, i.e. the (0) domain; e.g. $\mathbf{X}_T^\top \mathbf{X}_T = n_0 \mathbf{I}_p$.*) The KKT condition gives $\hat{\boldsymbol{\beta}}_{\text{bias}}^{(0)} = \frac{n_0 \hat{\boldsymbol{\beta}}_{\text{OLS}}^{(0)} + \lambda \hat{\boldsymbol{\beta}}^{(1)}}{n_0 + \lambda}$ — a weighted average of target OLS and the source anchor; subtracting $\boldsymbol{\beta}^{(0)*}$ gives the $\frac{\lambda}{\lambda + n_0}$ bias factor directly.

$$\mathbb{E} \|\hat{\boldsymbol{\beta}}_{\text{bias}}^{(0)} - \boldsymbol{\beta}^{(0)*}\|^2 = \underbrace{\left(\frac{\lambda}{\lambda + n_0}\right)^2 \|\boldsymbol{\delta}\|^2}_{\text{bias toward source}} + \underbrace{\frac{\sigma^2 n_0 p}{(\lambda + n_0)^2}}_{\text{shrunk variance}}.$$

- Bias term scales with $\|\boldsymbol{\delta}\|^2$ — “how far is source.”
- Variance term is shrunk by ridge — “how much do we trust source.”
- Optimal λ^* : set $\partial_\lambda \text{MSE} = 0$ (bias slope = variance slope), giving $\lambda^* = \sigma^2 p / \|\boldsymbol{\delta}\|^2$.

Theorem 4.2.1 (Kuzborskij and Orabona, 2017)

For squared loss with bounded covariates, the biased ridge estimator with anchor $\hat{\beta}^{(1)}$ trained on n_1 source samples satisfies, with high probability,

$$\mathbb{E}[R^{(0)}(\hat{\beta}_{\text{bias}}^{(0)})] - R^{(0)}(\beta^{(0)*}) \lesssim \min\left(\frac{p}{n_0}, \frac{p}{n_0 + n_1}\right) + \frac{\|\delta\|^2}{1 + n_0 \|\delta\|^2 / p}.$$

Two regimes:

- Small $\|\delta\|$: the rate is $\mathcal{O}(p/(n_0 + n_1))$ – effectively one big sample.
- Large $\|\delta\|$: the rate is $\mathcal{O}(p/n_0)$ – target-only fallback.

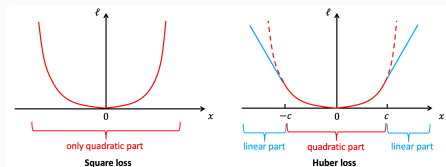
Outlier robustness via redescending loss

Replace squared loss with Huber-like redescending ρ :

$$\hat{\beta} = \arg \min \sum_i \rho(y_i - \mathbf{x}_i^\top \beta) + \lambda \|\beta - \hat{\beta}^{(1)}\|^2.$$

Theorem 4.2.2 (informal; redescending M-estimation, cf. She and Owen, 2011)

Under ϵ -corrupted target labels, a biased redescending estimator is expected to lose only a $\sqrt{\epsilon}$ corruption term, while the squared-loss biased ridge degrades linearly in ϵ .



Check-in: tuning λ

In the orthonormal-design example, the optimal $\lambda^ = \sigma^2 p / \|\delta\|^2$ requires knowing $\|\delta\|$.*

Which strategy gives an honest data-driven choice?

- (a) Plug in $\|\hat{\beta}_{\text{naive}}^{(0)} - \hat{\beta}^{(1)}\|$ as a noisy estimate of $\|\delta\|$.
- (b) Cross-validate λ on the target sample.
- (c) Use $\lambda = n_0$ always.
- (d) Both (a) and (b) are reasonable; (b) is more robust to high-variance plug-ins.

Answer

(d). In small target samples $\hat{\beta}_{\text{naive}}^{(0)}$ is itself very noisy, so plug-in for $\|\delta\|$ can be wildly off. Cross-validation directly minimizes target risk and is the modern practical choice – though it requires n_0 to be at least moderate.

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§4.2.3: Adaptive ℓ_2 -penalty (Duan et al.)

The adaptive estimator

Two-step procedure (Duan and Wang, 2022):

1. Compute $\hat{\beta}^{(1)}$ on source.
2. Use *data-driven* $\hat{\lambda}$ in $\hat{\beta}_{\text{bias}}^{(0)} = \arg \min \|\mathbf{y}_T - \mathbf{X}_T \beta\|^2 + \hat{\lambda} \|\beta - \hat{\beta}^{(1)}\|^2$, where $\hat{\lambda}$ minimizes a penalized empirical risk plug-in (or CV on target).

Adaptive theorem

Theorem 4.3.1 (Duan and Wang, 2022)

For Gaussian designs with subgaussian noise, the adaptive estimator $\hat{\beta}_{\text{adapt}}^{(0)}$ satisfies, with high probability,

$$\frac{1}{n_0} \|\mathbf{X}_T(\hat{\beta}_{\text{adapt}}^{(0)} - \beta^{(0)*})\|^2 \lesssim \frac{p}{n_0 + n_1} + \min\left(\frac{p}{n_0}, \|\delta\|^2\right).$$

Reading it: $p/(n_0 + n_1)$ is the pooled variance (always collected); $\min(p/n_0, \|\delta\|^2)$ because the bias cannot exceed target-only variance (p/n_0) nor the true squared drift ($\|\delta\|^2$) — adaptive $\hat{\lambda}$ picks the better branch.

- Achieves the better of the two regimes *without knowing* $\|\delta\|$.
- This is the precise sense in which biased regularization is “safe transfer.”

Check-in: when does $\hat{\beta}_{\text{adapt}}^{(0)}$ beat target-only?

Figure's setup ($p = 30$, $n_0 = 60 > p$, $\sigma = 1$): target-only OLS pays variance ≈ 1 ; transfer pays bias $\|\delta\|^2$.

- (a) It always beats target-only by a constant factor.
- (b) It beats target-only roughly when $\|\delta\|^2 \lesssim \text{target-only variance}$ ($\sim p/n_0$).
- (c) It only beats target-only when $\|\delta\| = 0$.
- (d) It never beats target-only.

Answer

(b). The adaptive bound's bias term is $\min(\text{var}_{\text{target}}, \|\delta\|^2)$: while the source bias $\|\delta\|^2$ stays below the variance you would otherwise pay, transfer wins; beyond that, adaptation falls back to the target-only rate. The next figure shows exactly this crossover.

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Numerical illustration: biased ridge

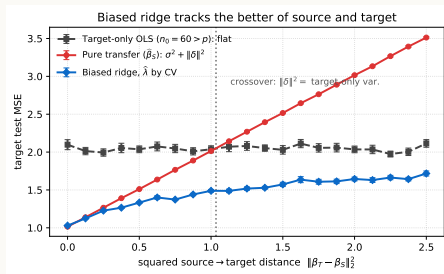
Numerical illustration: setup

Linear regression, $p = 30$, source size $n_1 = 2000$ (large), target size $n_0 = 60 > p$, noise $\sigma = 1$.

- Source coefficient $\beta^{(1)*}$ random; target $\beta^{(0)*} = \beta^{(1)*} + \delta$, sweeping $\|\delta\|^2 \in [0, 2.5]$.
- Three estimators: **target-only OLS** (since $n_0 > p$, it is flat in δ), **pure transfer** ($\hat{\beta}^{(1)}$ as-is), **biased ridge** with $\hat{\lambda}$ chosen by 5-fold CV.

Code: `sims/L4_biased_ridge.py`.

Numerical illustration: the safe-transfer crossover



- **Target-only OLS** (black): *flat* — $n_0 > p$, pays variance $\sim p/n_0$ regardless of drift.
- **Pure transfer** (red): rises as $\sigma^2 + \|\delta\|^2$ (bias = squared drift).
- They cross at $\|\delta\|^2 = \text{target-only variance}$ (dotted): borrow left, go target-only right.
- **Biased ridge**, $\hat{\lambda}$ by CV (blue): the *safe* lower curve — never worse than either.

Numerical takeaway

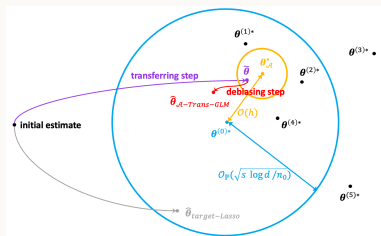
- Biased regularization gives the “safest” of three estimators in this experiment.
- With $\hat{\lambda}$ chosen by CV, the biased ridge stays below *both* pure transfer and target-only across the whole sweep — the empirical face of the Duan et al. adaptive bound: it attains the minimum-of-rates without knowing $\|\delta\|$.
- Same machinery extends to high dimensions next.

§4.3: High-dimensional extensions

ℓ_1 -penalty: TransLasso

Sparse regression in $p \gg n$. Idea (Li et al., 2021; Tian and Feng, 2023): generalize the biased- ℓ_2 trick to biased- ℓ_1 .

1. Estimate $\hat{\beta}^{(1)}$ on source (lasso).
2. Estimate the contrast $\hat{\delta} = \arg \min \|\mathbf{y}_T - \mathbf{X}_T(\hat{\beta}^{(1)} + \delta)\|^2 + \lambda_\delta \|\delta\|_1$ (the residual $\tilde{\mathbf{y}}_T = \mathbf{y}_T - \mathbf{X}_T \hat{\beta}^{(1)}$ isolates the small drift: $\hat{\delta} = \arg \min \|\tilde{\mathbf{y}}_T - \mathbf{X}_T \delta\|^2 + \lambda_\delta \|\delta\|_1$).
3. Output $\hat{\beta}_{\text{trans}}^{(0)} = \hat{\beta}^{(1)} + \hat{\delta}$.



Software: our group's R/Python packages implement TransLasso/TransGLM —
https://yangfengstat.github.io/projects/transfer_learning/.

TransLasso theory (informative source)

Theorem 4.5.1 (Li et al., 2021; Tian and Feng, 2023)

Suppose $\beta^{(1)*}, \delta$ are s -sparse and h -sparse, source size $n_1 \gg n_0$, designs satisfy RE. Then with high probability,

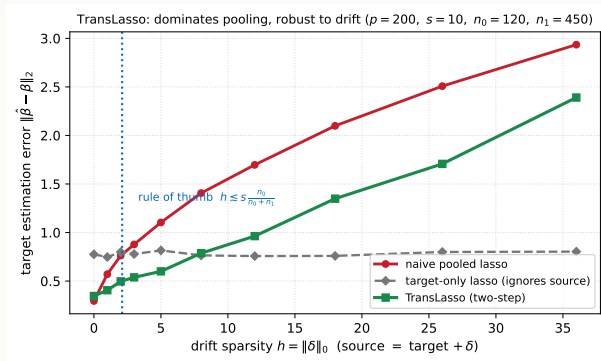
$$\frac{1}{n_0} \|\mathbf{X}_T(\hat{\beta}_{\text{trans}}^{(0)} - \beta^{(0)*})\|^2 \lesssim \frac{s \log p}{n_0 + n_1} + \frac{h \log p}{n_0}.$$

Compared to a target-only lasso rate of $\frac{s \log p}{n_0}$:

- $h \ll s$: rate $\approx s \log p / (n_0 + n_1)$ — linear gain in the source size.
- $h \sim s$: source as complex as the drift, no gain. Rule of thumb: benefit needs $h \lesssim s n_0 / (n_0 + n_1)$.

TransLasso vs. drift sparsity: the crossover

Sweep the drift sparsity $h = \|\delta\|_0$ (source = target + δ); compare target estimation error.



- **Naive pooling** wins at $h=0$ but degrades fast, crossing target-only near $h \approx s n_0 / (n_0 + n_1)$.
- **TransLasso** dominates pooling *everywhere*, beating target-only out to $h \approx s n_1 / (n_0 + n_1)$.

L4.translasso.py

When the informative-source set S is unknown

In practice, only *some* of the available sources are useful. Two strategies (Tian and Feng, 2023):

- **Source selection.** Estimate which sources have small $\|\boldsymbol{\delta}_k\|_1$; run TransLasso only on the selected subset.
- **Aggregation.** Average individual TransLasso estimators with stability-selection-style weights.

Either approach achieves the oracle rate up to a logarithmic penalty for uncertainty about S .

Mis-specification: GLMs with model error

Real source models are never exactly the target. Letting $\beta^{(1)*}$ be the best linear approximation of the source GLM, results extend (Tian and Feng, 2023; Bastani, 2021):

Practical message: a slightly wrong source model is much better than a tiny target sample.

- Add an irreducible mis-specification term to the bound.
- As long as mis-specification $\ll \sqrt{s \log p / n_0}$, transfer remains beneficial.

§4.3.3: Block penalty with multi-task learning

Multi-task learning: shared sparsity

So far one source feeds one target via a single contrast δ ; multi-task learning generalizes to K symmetric tasks that borrow strength through shared support rather than a designated anchor.

K related tasks, each with $\beta^{(k)} \in \mathbb{R}^p$. Stack them as columns of $B = [\beta^{(1)} \mid \dots \mid \beta^{(K)}] \in \mathbb{R}^{p \times K}$.

- Naive: lasso on each task separately.
- Block-sparsity prior: many features are zero *across all tasks simultaneously*.
- Encode this with a row-norm penalty:

$$\hat{B} = \arg \min_B \sum_{k=1}^K \|\mathbf{y}^{(k)} - \mathbf{X}^{(k)} \beta^{(k)}\|^2 + \lambda \sum_{j=1}^p \|B_{j,\cdot}\|_q.$$

Choices for q : $q = 2$ gives **group lasso**; $q = \infty$ gives the ℓ_1/ℓ_∞ norm; mixtures give “dirty” variants.

Group-lasso theorem

Theorem 4.6.1 (Lounici et al., 2011)

For Gaussian designs satisfying group-RE, group lasso with $\lambda \asymp \sigma \sqrt{(\log p + K)/n}$ achieves

$$\frac{1}{nK} \sum_k \|\mathbf{X}^{(k)}(\hat{\boldsymbol{\beta}}^{(k)} - \boldsymbol{\beta}^{(k)*})\|^2 \lesssim \frac{s(K + \log p)}{nK},$$

where s is the number of nonzero rows of B^ .*

- Compared to single-task lasso rate $s \log p/n$, the multi-task gain is roughly $K/(K + \log p)$ – nontrivial when K large.
- Information sharing happens through the shared support, not through the coefficient values.

Theorem 4.6.2 (Negahban and Wainwright, 2011)

Under similar conditions, the ℓ_1/ℓ_∞ MTL estimator achieves a rate of order $\frac{s(K+\log p)}{nK}$ plus an irreducible term proportional to the row-wise dispersion of B^ .*

- Best when coefficients are not just shared-support but *shared-magnitude* across tasks.
- Vulnerable to a single “outlier” task; this motivates the dirty model.

Dirty model: $B = S + O$ decomposition

Theorem 4.6.3 (Jalali et al., 2010, 2013)

Decompose $B = S + O$ where S is row-sparse and O is element-sparse. The dirty estimator $\min \frac{1}{2} \|\mathbf{y} - \mathbf{X}(S + O)\|^2 + \lambda_S \|S\|_{1,\infty} + \lambda_O \|O\|_1$ matches the better of the row-wise rate and the element-wise rate.

- Adapts to the unknown mixture of shared and task-specific support.
- The MTL analogue of the adaptive penalty selection from §4.2.3.

Check-in: which penalty?

Six cancer cohorts; gene-expression covariates ($p = 20,000$). Ten genes are causal across all cohorts; each cohort has 1–3 idiosyncratic markers. Which estimator best matches this structure?

- (a) Lasso applied independently to each cohort.
- (b) Group lasso (ℓ_1/ℓ_2): forces shared support.
- (c) ℓ_1/ℓ_∞ : forces shared magnitudes.
- (d) Dirty model: row-sparse S + element-sparse O .

Answer

(d). The 10 shared causal genes correspond to row-sparse S ; the 1–3 idiosyncratic markers per cohort correspond to element-sparse O . The dirty model exactly captures both. Pure group lasso (b) misses the idiosyncratic markers; ℓ_1/ℓ_∞ (c) wrongly forces equal magnitudes.

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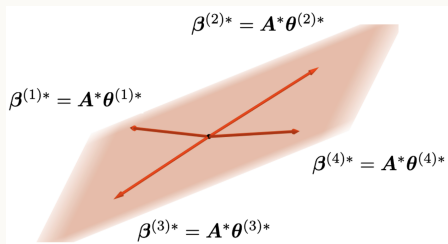
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§4.3.4: Shared low-dimensional representations

From shared support to shared subspace

Block penalties share the *same features* (support). A more flexible assumption: tasks share a low-dimensional *linear representation*.

- K tasks, each $\beta^{(k)} = \mathbf{A} \boldsymbol{\theta}^{(k)}$ with a *shared* $\mathbf{A} \in \mathbb{R}^{p \times r}$, $r \ll p$, and task-specific $\boldsymbol{\theta}^{(k)} \in \mathbb{R}^r$.
- Information now flows through a shared *subspace* (the column space of $\mathbf{A}$), not a shared support.
- Only r directions matter — but *which* directions is unknown.



Adaptivity, minimaxity, robustness

T tasks, size n , ambient p , representation dim $r \ll p$, similarity h , outlier fraction ϵ_{out} (all unknown).

Theorem 4.7.1 (Tian et al., 2025, simplified)

The adaptive spectral estimator attains, for each inlier task t ,

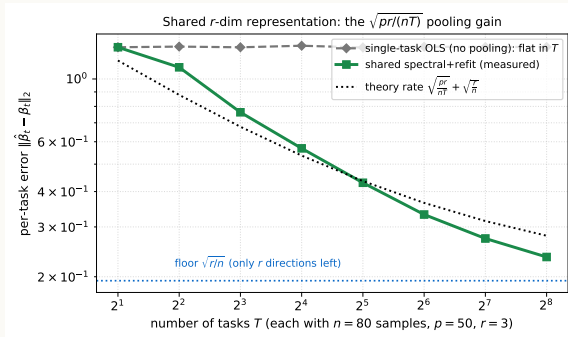
$$\|\hat{\beta}^{(t)} - \beta^{(t)*}\|_2 \lesssim \underbrace{\left(\sqrt{\frac{pr}{nT}}\right)}_{\text{pool}} + \underbrace{\sqrt{\frac{r}{n}}}_{\text{task}} + \underbrace{h}_{\text{sim.}} + \underbrace{\sqrt{r \epsilon_{\text{out}}}}_{\text{outlier}} \wedge \sqrt{\frac{p}{n}},$$

*and the rate is **minimax-optimal** (matching lower bound at $\epsilon_{\text{out}} = 0$).*

- Pooling across T tasks with a shared r -dim subspace: each of T tasks gives n equations about the *same* $\mathbf{A}$, so the effective sample is nT and ambient p drops to r — a $\sqrt{pT/r}$ -type gain over target-only.
- Adapts to unknown h ; the $\wedge \sqrt{p/n}$ is the single-task safety net (no negative transfer).
Full statement + rate table: backup appendix.

The pooling gain $\sqrt{pr/(nT)}$, illustrated

T tasks, each $\beta^{(t)} = \mathbf{A} \boldsymbol{\theta}^{(t)}$, shared $\mathbf{A} \in \mathbb{R}^{p \times r}$. Estimate $\mathbf{A}$ from the per-task fits (top- r subspace), then refit each task inside it.

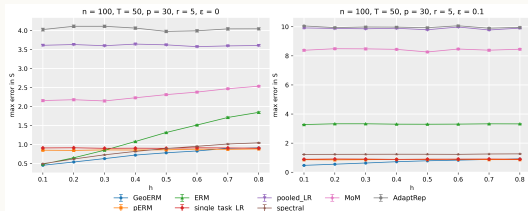


- **Single-task** OLS learns all p coordinates from n points — flat in T (no borrowing).
- **Shared** falls as $\sqrt{pr/(nT)} + \sqrt{r/n}$: more tasks sharpen $\mathbf{A}$, amortizing the ambient- p cost down to the r -dim floor.

Geometry-aware extension: GeoERM

Orthonormal $\mathbf{A}$ lives on a *Stiefel / Grassmann manifold*; a flat $\mathbb{R}^{p \times r}$ gradient step ignores that curvature.

- A flat update on $\mathbf{A}$ can fall off the orthonormality constraint, losing the shared subspace; **GeoERM** (Chen and Feng, 2026) keeps $\mathbf{A}$ on the Stiefel manifold (Riemannian gradient + retraction), so the representation stays consistent across updates.
- **Empirics:** HAR 0.98% vs. 1.33% best baseline; MNIST 6.4% vs. 28.9%; $\sim 15\text{--}40\times$ faster.
- **Theory:** a statistical rate on the manifold is *open*.



(§4.4–§4.5): drop as a unit if running short on time =====

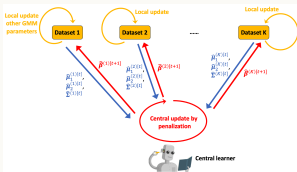
§4.4: Unsupervised federated transfer

Transfer without labels, without a central pool

Two frontier extensions, each dropping an assumption: no labels (EM, §4.4) and a privacy budget (§4.5) — same thread, still bias toward shared structure.

So far: labeled regression. Now *unsupervised* transfer.

- K clients, each with *unlabeled* data from a mixture (e.g. a Gaussian mixture); related but heterogeneous across clients.
- **Federated constraint:** raw data cannot leave a client; only model parameters / sufficient statistics may be communicated.
- Goal: borrow strength across clients to estimate the shared mixture structure better than single-client EM.



Federated EM

K clients, size n , dim d , R mixture components, similarity h , outlier fraction $\epsilon_{\text{out}} < \frac{1}{3}$.

Theorem 4.8.1 (Tian et al., 2024, GMMs, simplified)

After $T \gtrsim \log n$ rounds, federated EM satisfies

$$\max_k \text{err}_k \lesssim \underbrace{\kappa_0^T}_{\text{iter.}} + \underbrace{R^2 \sqrt{\frac{d}{nK}}}_{\text{federated}} + \underbrace{R^2 \sqrt{\frac{1}{n}}}_{\text{mixing}} + \underbrace{\min\{h, R^2 \sqrt{\frac{d}{n}}\}}_{\text{heterog.}} + \underbrace{\epsilon_{\text{out}} R^2 \sqrt{\frac{d}{n}}}_{\text{outlier}},$$

with contraction $\kappa_0 \in (0, 1)$.

- $\sqrt{K}$ **gain**: averaging sufficient statistics from K clients pools nK samples for the shared mean, so noise is $\sqrt{d/(nK)}$ not $\sqrt{d/n}$; $\min\{h, \cdot\}$ ensures no client is hurt.
- Only sufficient statistics (means, weights) are shared, not raw data. Hard part: GMM components have no canonical order, so the server must *match* components across clients before averaging (else it averages apples and oranges).

Centralized counterpart: robust multi-task EM on GMMs

Centralized sibling (Tian et al., 2026); unsupervised $\Rightarrow$ components are identified only up to a per-task **label permutation** (alignment is the key obstacle).

Theorem 4.8.2 (simplified, minimax)

Excess mis-clustering error for each inlier task k :

$$R(\widehat{\mathcal{C}}^{(k)}) - R^* \lesssim \underbrace{\frac{p}{n_S}}_{\text{oracle}} + \underbrace{\frac{\log K}{n_k}}_{\text{scalars}} + \underbrace{h^2 \wedge \frac{p + \log K}{n_k}}_{\text{similarity}} + \underbrace{\epsilon_{\text{out}}^2 \frac{p + \log K}{\max_k n_k}}_{\text{outlier}},$$

where $n_S = \sum_{k \in S} n_k$.

- Factor- K gain over single-task EM; rate $\leq (p + \log K)/n_k$ always $\Rightarrow$ never worse.

§4.5: Federated transfer + differential privacy

Transfer under a privacy budget

Federated transfer where each client must *protect* its data.

- Privacy model: (ϵ, δ) -**federated DP** — enforced *locally at each site*, no trusted central server (strictly between central and local DP).
- Calibrated noise injected into shared statistics; raw data and exact gradients never leave.
- Question: how much does privacy *cost* the transfer estimator?

(δ here is the DP failure probability — unrelated to the drift vector δ above.)

Privacy–utility trade-off

Low-dim regression, balanced $n_k = n$, K sources, dim d , $h = 0$:

$$\text{error} \asymp \underbrace{\sqrt{\frac{d}{nK}}}_{\text{non-private}} + \underbrace{\frac{d}{n\varepsilon\sqrt{K}}}_{\text{privacy cost}}.$$

Non-private	$\sqrt{d/(nK)}$
Central DP	$\sqrt{d/(nK)} + d/(nK\varepsilon)$
Federated DP	$\sqrt{d/(nK)} + d/(n\varepsilon\sqrt{K})$
Local DP	$d/(\sqrt{nK}\varepsilon)$

Federated DP sits between the extremes: each client adds noise locally (worse than central's single injection), but the K -client sum still averages noise down by $\sqrt{K}$ (better than local DP, which never pools). Transfer helps while $n\varepsilon^2/d \gtrsim 1$ (Li et al., 2024).

Lecture summary

Lecture summary

Posterior drift requires *joint* use of source and target data; reweighting is useless.

Biased regularization delivers safe transfer in escalating settings:

- Linear / GLM regression $\Rightarrow$ biased ℓ_2 , with adaptive $\hat{\lambda}$ giving min-rate.
- High-dim $\ell_1 \Rightarrow$ TransLasso; gain $\propto n_1/(n_0 + n_1)$ when source is informative.
- Multi-task with shared sparsity $\Rightarrow$ group lasso, ℓ_1/ℓ_∞ , dirty models.
- Shared low-dim *representation* $\Rightarrow$ adaptive, minimax, robust subspace transfer (§4.3.4); GeoERM adds geometry.

Beyond today: federated and differentially private transfer (*time permitting*).

Across all: bias toward shared source structure, calibrate from the target.

Course summary: three threads, one picture

- Lecture 2: divergence-style bounds tell you *whether* transfer is possible.
- Lecture 3: covariate shift $\Rightarrow$ density-ratio reweighting recovers oracle target risk.
- Lecture 4: posterior drift $\Rightarrow$ biased regularization gives safe transfer with adaptive penalty.

Different shifts call for different tools; the unifying theme is that the source becomes useful exactly when explicit statistical structure (small divergence, bounded ratio, small parameter contrast) lets it.

Recent frontiers: shared-representation transfer (adaptive / minimax / robust; GeoERM), and unsupervised, federated, and differentially private transfer.

Thank you.

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Backup A1 — Shared representations (full) i

Model. $\beta^{(t)*} = \mathbf{A}^{(t)*} \boldsymbol{\theta}^{(t)*}$, $\mathbf{A}^{(t)*} \in \mathcal{O}^{p \times r}$; inlier set S with $\max_{t \in S} \|\mathbf{A}^{(t)*} \mathbf{A}^{(t)*\top} - \bar{\mathbf{A}} \bar{\mathbf{A}}^\top\|_2 \leq h$; outlier fraction $\epsilon_{\text{out}} = |S^c|/T$.

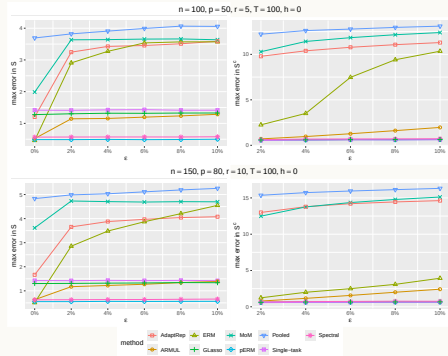
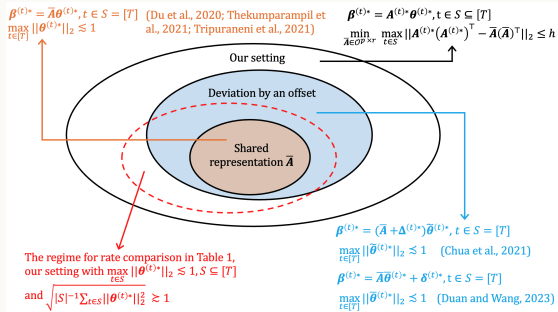
Theorem 4.12.1 (Spectral method, Tian–Gu–Feng 2025, Thm. for $\zeta^{(t)} \asymp \bar{\zeta} \asymp 1$)

$$\|\widehat{\beta}^{(t)} - \beta^{(t)*}\|_2 \lesssim \left(\sqrt{\frac{pr}{nT}} + \sqrt{\frac{r}{n}} + h + \sqrt{r \epsilon_{\text{out}}} \right) \wedge \sqrt{\frac{p}{n}}, \quad \forall t \in S,$$

minimax-optimal at $\epsilon_{\text{out}} = 0$ (matching lower bound, $O(1)$ condition number).

Assumptions. sub-Gaussian design ($0 < c \leq \lambda_{\min} \leq \lambda_{\max} \leq C$); task diversity $\sigma_r(\mathbf{B}_S^* / \sqrt{|S|}) \gtrsim \bar{\zeta} / \sqrt{r}$; $n \gtrsim p + \log T$. Robustness: outlier tasks themselves estimated at single-task rate $\sqrt{(p + \log T)/n}$.

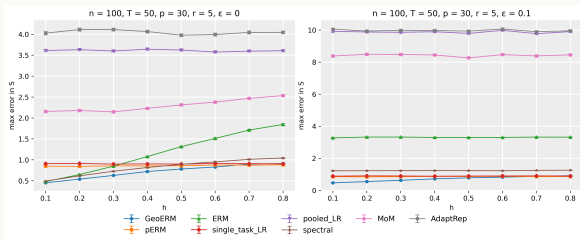
Backup A1 — Shared representations (full) ii



Backup A2 — GeoERM (convergence + experiments)

Guarantee (optimization). Riemannian gradient descent on $\text{St}(p, r)$ with Lipschitz $\tilde{\nabla}f$ and small step α satisfies a descent inequality $f(x_{k+1}) \leq f(x_k) - c\alpha\|\tilde{\nabla}f(x_k)\|^2$ (Absil et al. 2008, Thm. 4.3.1); every accumulation point is first-order stationary. **A statistical rate is open** (Riemannian concentration inequalities are future work).

- Per-iteration subspace penalty cost $O(p^3) \rightarrow O(pr^2)$ (Stiefel constraint).
- HAR ($r=10$): $0.98\% \pm 0.15$ vs. pERM $1.33\% \pm 0.19$; ~ 54 s vs. 777s ($\sim 15\times$).
- MNIST-C (45T): 6.38% vs. pERM 28.9% vs. ERM 50.1%; robust to adversarial label-shuffled tasks.



Backup A3 — Federated EM / FedGrEM (full) i

FedGrEM. Each client runs a local E-step + gradient M-step; the server aggregates with a group- ℓ_2 penalty shrinking each client to a common center (robust to outliers).

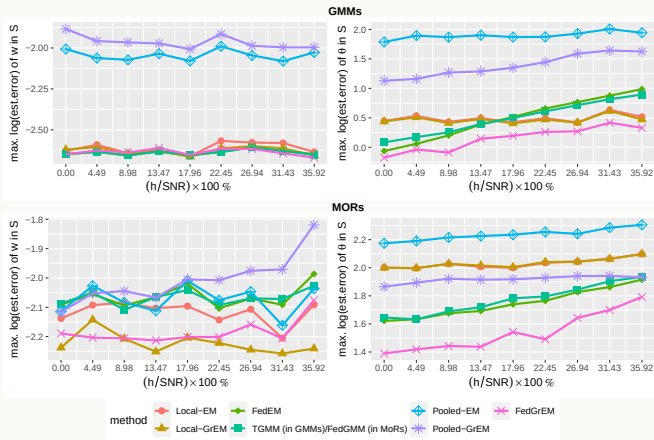
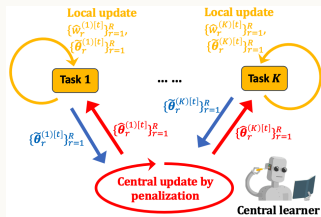
Theorem 4.12.2 (Tian–Weng–Feng 2024, GMM corollary)

For $\epsilon_{\text{out}} = |S^c|/K < 1/3$, w.p. $1 - o(1)$, for all $T \geq 1$,

$$\max_{k \in S, r} \left(|\hat{w}_r^{(k)[T]} - w_r^{(k)*}| \vee \|\hat{\theta}_r^{(k)[T]} - \theta_r^{(k)*}\|_2 \right) = \tilde{O} \left(\kappa_0^T + R^2 \sqrt{\frac{d}{nK}} + R^2 \sqrt{\frac{1}{n}} + \min\{h, R^2 \sqrt{\frac{d}{n}}\} + \epsilon_{\text{out}} R^2 \sqrt{\frac{d}{n}} \right).$$

Assumptions. init within $O(\Delta)$; separation $\Delta \gtrsim \log R$, $n \gtrsim R^4[d + \log(RK)]\Delta^{-2}$; population contraction $\kappa \in (0, 1)$ ($\kappa \asymp R^2 e^{-C\Delta^2}$). FedGrEM is the ICML 2024 paper.

Backup A3 — Federated EM / FedGrEM (full) ii



Backup A4 — Robust MTL/TL on GMMs (full)

Setup. K tasks, binary GMMs; discriminant $\beta^{(k)} = (\Sigma^{(k)})^{-1}(\mu_2^{(k)} - \mu_1^{(k)})$; inliers S with $\max_{k \in S} \|\beta^{(k)} - \bar{\beta}\|_2 \leq h$; penalized multi-task EM with a free center.

Theorem 4.12.3 (Tian–Weng–Xia–Feng 2026, upper + lower)

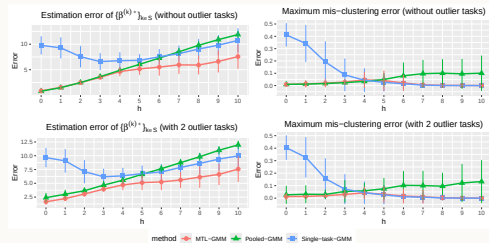
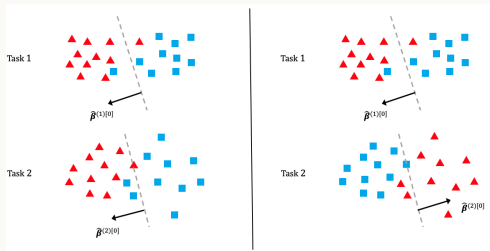
For inlier k , excess mis-clustering error

$$R(\widehat{C}^{(k)}) - R^\star \lesssim \frac{p}{n_S} + \frac{\log K}{n_k} + \left(h^2 \wedge \frac{p + \log K}{n_k} \right) + \epsilon_{\text{out}}^2 \frac{p + \log K}{\max_k n_k},$$

and a matching minimax lower bound holds on terms (I)–(III).

Backup A4 — Robust MTL/TL on GMMs (cont.)

Assumptions. $\min_{k \in S} n_k \gtrsim p + \log K$; Mahalanobis SNR $\Delta^{(k)} \geq C$; label-aligned initialization. TL version shrinks the target toward the learned center.



Backup A5 — Federated transfer + DP (full) i

Federated DP. Each site applies its own (ε, δ) -DP mechanism (clip + Gaussian noise) before communicating; no trusted server. Targets: mean, low-/high-dim regression, M-estimation.

Theorem 4.12.4 (Li–Tian–Feng–Yu, low-dim regression)

$$\|\tilde{\beta} - \beta\|_2 \lesssim r_{\text{target}} \wedge \left(h + \log^2(N/\eta) \sqrt{\frac{d \log(1/\delta)}{\sum_{k \in \{0\} \cup \mathcal{A}} \min\{n_k, n_k^2 \varepsilon^2 / d\}}} \right).$$

Balanced $n_k = n$, $h = 0$: cost interpolates — central DP $d/(nK\varepsilon)$, **FDP** $d/(n\varepsilon\sqrt{K})$, local DP $d/(\sqrt{nK}\varepsilon)$.

- Privacy is free when $n\varepsilon^2/d \gtrsim 1$; source detection prevents negative transfer.
- Also: mean rate $1/\sqrt{n_0} + 1/(n_0\varepsilon)$; high-dim sparse rate with $s \log d$ factors.

Backup A5 — Federated transfer + DP (full) ii

